

SEAN VINCENT VIEN V. VIÑAS

Computer Engineer | AI Engineer

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SUMMARY

A detail-focused and result-driven Computer Engineer with several experiences with Artificial Intelligence (AI), Machine Learning (ML), Internet of Things (IoT) architecture, and Edge Computing. Demonstrates strong academic excellence alongside hands-on leadership experience managing student organization finances, large-scale campus events, and cross-functional technical projects. Seeking to leverage technical expertise in software engineering, embedded systems, and machine learning deployment.

EDUCATIONAL BACKGROUND

UNIVERSITY OF THE EAST | MANILA

Recto Ave., Manila

Bachelor of Science in Computer Engineering (BSCpE)

June 2026

- **Relevant Courses:** Data Structures and Algorithms, Database Management System, Object Oriented Programming, Embedded Systems, Internet of Things, Machine Learning, Artificial Intelligence
- **Graduated:** Magna Cum Laude | GWA: 1.32
- **Recognition:** College Distinguished Graduate | Leadership Awardee for Men

MACHINE LEARNING & IoT PROJECTS

SENTRYSCAN | *Personal Project* | Developer | 2026

Human Detection and Tracking System for Real-Time Zone Monitoring and Dwell-Time Analytics.

- Built an end-to-end human tracking pipeline using **YOLOv8** and **BoT-SORT**, with **homography-based ground-plane mapping** for accurate zone detection on angled cameras.
- Engineered zone entry/exit logging and per-person dwell-time computation, with trajectory visualization on the output video.
- Developed a full-stack web app (**FastAPI**, **Next.js**) with a REST API for video upload, interactive zone drawing, and playback synced to a live entry/exit log.

AIXIA | *Personal AI Assistant Project* | Developer | 2026

A Full-Stack Retrieval-Augmented Generation (RAG) Chatbot.

- Designed and developed a full-stack RAG chatbot using **Django REST Framework**, **LangChain**, **ChromaDB**, **PostgreSQL**, and **Next.js**.
- Applied professional-grade software discipline — structured documentation, QA practices, and incremental testing.
- Self-hosted on a **Raspberry Pi 5** via Cloudflare Tunnel, cutting infrastructure costs to a domain-only annual expense.

THE SMOKi PROJECT | *Design Project* | Project Head / Product Owner | 2026

Design and Development of an IoT-Driven Vehicle Smoke Emission Classification System using Convolutional Neural Networks with Data Analytics Management.

- Developed an end-to-end computer vision pipeline for vehicle smoke emission classification using a **CNN-based model (YOLOv8)**, deployed for Edge Computing on **Raspberry Pi 5 + HAILO-8**.
- Gathered, labeled, and preprocessed image and environmental datasets from the IoT device; trained and evaluated the model using **Python**, **PyTorch**, and **Scikit-learn**.
- Engineered a machine learning model capable of classifying smoke color (Black/White) and integrated analytics dashboards with privacy-preserving image processing.

THE AEROBAND PROJECT | *Design Project* | Project Head / Product Owner | 2025

An IoT-Powered Wearable for Real-Time Air Quality Monitoring and Pollution Detection.

- Designed and prototyped a wearable IoT device for real-time air quality monitoring using Arduino R4.
- Utilized Air Quality Sensors: BME680, MiCS-6814, and PMS7003.
- Processed gas and particulate sensor data for pollution detection and trend analysis.
- Applied data cleaning and exploratory data analysis techniques to environmental dataset

TECHNICAL SKILLS

ML / AI STACK: Supervised Learning (Logistic Regression, Random Forest, Decision Tree, K-NN, Gradient Boosting), Computer Vision (CNN, YOLO), Model Training & Evaluation, LangChain, RAG, and LLMs.

PROGRAMMING & DATA ANALYTICS: Python, SQL, C++ (IoT & Embedded Systems)

LIBRARIES & FRAMEWORKS: Git/Github, Scikit-learn, Numpy, PyTorch, Linux Fundamentals

DATABASE: MySQL, VectorDB, ChromaDB

PROFESSIONAL AND MANAGEMENT-RELEVANT: Google Suite, Scrum, Jira

AGILE & PRODUCT: Scrum, backlog refinement, requirements gathering, user-story translation, product scoping, roadmap alignment, stakeholder communication

WORK EXPERIENCE

AUTOMATE IT SOLUTIONS | *Junior AI Developer*

Pasig City, Metro Manila (WFH)

Full-time / Permanent

August 2026 - Present

- Design, develop, test, and maintain **AI-enabled applications, automations**, and internal software tools, from dataset preparation through deployment.
- Integrate AI models and services through approved **APIs, frameworks, databases**, and **cloud platforms**, writing secure and maintainable code.
- Evaluate model and application outputs for accuracy, reliability, safety, latency, and cost, and document findings for the team.

HIGHLY SUCCEED INC. | *Internship*

Mandaluyong City, Metro Manila

Project Management and Quality Assurance Intern

June - August 2025

- **Quality Assurance:** Improved project reliability by performing end-to-end testing and managing a detailed bug-tracking lifecycle in Jira, utilizing Excel to document and report on quality metrics.
- **Project Management:** Assisted in cross-functional collaboration on system validation and documentation and maintaining rigorous meeting records to align project goals

LEADERSHIP EXPERIENCE

UE COLLEGE OF ENGINEERING | *Department of Computer Engineering*

SMX Convention Center

University Representative | Project Head

September 2025

- **Duty:** Applied advanced engineering principles and technical problem-solving skills to successfully compete at a multi-university level.
- **Duty:** Represented the university in various external technical and academic competitions against other institutions.

ENGINEERING STUDENT COUNCIL (UE-ENSC)

UE - Manila

Business Manager

S.Y. 2025 - 2026

- **Duty:** Responsible for gathering sponsors on every event and outside negotiation with other entities that will benefit the engineering student body

SOCIETY OF COMPUTER ENGINEERING STUDENTS (SCpES)

UE - Manila

Vice President for Workshop and Trainings

S.Y. 2025 - 2026

- Facilitated hands-on training sessions in Emerging Technologies, including ML, IoT, Embedded Systems, and 3D Printing.
- **Circuit Minds (Project Head)**: Led curation of technical quiz questions and venue logistics for a CpE quiz bee competition.
- **Gotta GIT Advanced! (Project Head)**: Led curriculum design and technical delivery for an advanced Git training workshop.
- **3D in Theory! (Project Head)**: Directed the technical flow of a 3D modeling seminar.
- **Digital to Physical (Project Head)**: Managed end-to-end execution of a 3D modeling and printing workshop, covering slicing optimization, material selection, and post-processing.

UE COMMISSION OF ELECTIONS (UE-COMELEC)

UE - Manila

Student Commissioner for CENG

S.Y.2024 - 2025

- **Duty**: Managed election logistics, enforced electoral guidelines, and monitored the voting process within the engineering college.

SEMINARS CONDUCTED

ARDUINO 101 | Seminar and Workshop

UE - Manila

Resource Speaker

April 24, 2025

- Conducted a seminar and workshop teaching First and Second-Year students the basics of embedded systems using Arduino UNO and various sensors.

DATA TO DECISIONS | Seminar

De La Salle Araneta University

Resource Speaker

June 10, 2026

- Conducted a seminar teaching CpE and CS students the basics of machine learning, particularly, Supervised, Unsupervised, and Reinforced Learning through presentations.

AWARDS AND RECOGNITIONS

MAGNA CUM LAUDE | College Graduation

PICC - Pasay City

Silver Medal

June 09, 2026

COLLEGE DISTINGUISHED GRADUATE AWARDEE

PICC - Pasay City

Gold Medal

June 09, 2026

LEADERSHIP AWARDEE FOR MEN

PICC - Pasay City

Plaque

June 09, 2026

ACADEMIC EXCELLENCE AWARDEE

UE - Manila

Certificate of Recognition

S.Y. 2025 - 2026

EXEMPLARY PERFORMANCE IN OFF-CAMPUS COMPETITION

UE - Manila

Certificate of Recognition

S.Y. 2025 - 2026

1ST RUNNER-UP, PACKETHACKS X HACKTHECLIMATE HACKATHON

SMX Convention Center

Certificate of Recognition & Trophy

September 11, 2025

TOP 10 MOST OUTSTANDING COMPUTER ENGINEERING STUDENTS

UE - Manila

Certificate of Recognition

S.Y. 2023 - 2026

SPECIAL MERIT AWARD FOR OUTSTANDING INNOVATION

UE - Manila

Certificate of Recognition

S.Y. 2024 - 2025

SPECIAL MERIT AWARD FOR EXCELLENCE IN ACADEMIC AND SCHOLARLY OUTPUT

UE - Manila

Certificate of Recognition

S.Y. 2024 - 2025

DISTINGUISHED MERIT AWARD FOR EXCELLENCE

UE - Manila

Certificate of Recognition

S.Y. 2024 - 2025

CERTIFICATIONS

MACHINE LEARNING FOR BEGINNERS	Sololearn.com
<i>Course Certificate</i>	<i>2026</i>
CLAUDE 101	Claude Academy
<i>Course Certificate</i>	<i>2026</i>
PYTHON DEVELOPER	Sololearn.com
<i>Course Certificate</i>	<i>2024</i>
SCRUM FOUNDATION PROFESSIONAL CERTIFICATION (SFPC™)	CertiProf
<i>Course Certificate</i>	<i>2024</i>

CONTACT INFORMATION

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REFERENCES

References can be provided upon request.

NOTHING FOLLOWS